Alexander Mulcahy

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EDUCATION

University of Oxford — Oxford, England
BA Computer Science — 2024 – 2027
Expected 1st
Awarded multiple prizes for preliminary exam results
87% in Concurrent Programming

Exeter College — Exeter, England
Further Maths (A*), Maths (A*), Music (A*), Computer Science (A) — 2022 – 2024
Received the Faculty Award for Music
Achieved **72/75** on Computer Science NEA Project

WORK EXPERIENCE

Squid Software — Oxford, England
Data Visualization Intern — Feb 2026 – May 2026

- Achieved **1st** place out of 20 teams in the Oxford Computer Science Group Design Project
- Collaborated in a team to develop a web-based data visualization tool for a barge logistics company
- Applied Python for data processing and analysis, delivering actionable insights on a platform driven by React

Quill Project — Oxford, England
Web Development Intern — Dec 2025

- Designed a new staff page for quillproject.net and implemented with Bootstrap and jQuery
- Gathered user requirements and iterated design using Figma web design tools

Morrisons — Exeter, England
Delivery Driver — Jun 2025 – Oct 2025

- Delivered to up to 25 customers per day; Managed multiple orders at once; Ensured friendly service

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PROJECT EXPERIENCE

***Project*: Machine Learning Models**
Python / TensorFlow / SciKit-Learn

- Implemented a Naïve Bayes Classifier and a Convolutional Neural Network during my ML course
- The CNN could classify handwritten digits with 98% accuracy. Developed with SciKit-Learn library

***Project:* Minimax-Driven Connect4 CPU opponent**
Python

- Recursive algorithm implementing Minimax decision rule from game theory to create a computer opponent
- Adjustable difficulty by altering search depth, with ability to beat human player without significant processing time

***Project:* Games Website**
React / Python / Flask / C

- Designed and created a multi-user platform hosting a daily 5x5 Mini crossword at sandys.games
- Involved full-stack development, and optimizing a recursive algorithm to find valid crossword configurations in C

SKILLS AND INTERESTS

- Programming Languages: Python, JavaScript, Scala, Haskell, SQL
- Music: Grade 8 Jazz Piano, Devon Youth Jazz Orchestra (2022-2023)
- Sport: Ultimate Frisbee U17 National Champion (2022)